

Ch. 2 - General Theory of Experimental Modal Analysis

detailed handwritten-style notes - equations that run the lab

Whole chapter in 1 line:

Everything measured (FRF) = **POLES** (where modes live: damping + frequency) + **RESIDUES** (how strongly each mode shows at each point). Curvefitting = hunting these two. Ch.1 pictures become Ch.2 machines.

How to use:

Read S0, then 2.2 slowly (SDOF is the alphabet), then 2.3 (MDOF = teams of SDOFs). Do every [TRY] and worked example EX1-EX7 with a calculator. End with the self-test!

Legend: [KEY] key [LAB] lab [TRAP] trap [LINK] bridge [TRY]/EX do it

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11 Sep 2026



(0) Primer - Ch.1 pictures become Ch.2 machines

Ch.1 picture	Ch.2 machine
FRF peaks at natural frequencies	POLES $s = -z \omega_n + j \omega_d$ (damping + freq)
Imag-peak heights trace shape	RESIDUES $A_{ijr} \sim \text{shape-}i \times \text{shape-}j$
One row/column is enough	PROVEN by transfer-matrix symmetry
FRF = hidden SDOFs summed	DERIVED: $h_{ij} = \text{sum over modes}$
Curvefitting finds f, z , shapes	Curvefitting finds POLES + RESIDUES

Warm-up:

[TRY] Write your SDOF equation from memory: $m \ddot{x} + c \dot{x} + k x = f(t)$. Every 2.2 equation is this one in costume; every 2.3 equation is a TEAM of these, uncoupled by a clever coordinate change.

(2.1) Introduction - two models, one truth

	FEM (analytical)	EMA (experimental)
Starts from	Assumed M, K layout	Measured force + response
Gives	Freqs + shapes via EIGENSOLUTION	f, z , shapes via CURVEFITTING
Strength	Design-stage what-ifs, no hardware	Ground truth of real structure
Weakness	Wrong assumptions = wrong modes	Bad measurements = wrong params

The loop that matters:

[KEY] Build FEM $\rightarrow$ predict $\rightarrow$ TEST $\rightarrow$ **verify/correct FEM with modal data** $\rightarrow$ trusted model. EMA boomed since the 1980s (cheap analyzers, MIMO, better estimators, OMA). Ch.2 is the theory core making it legitimate.

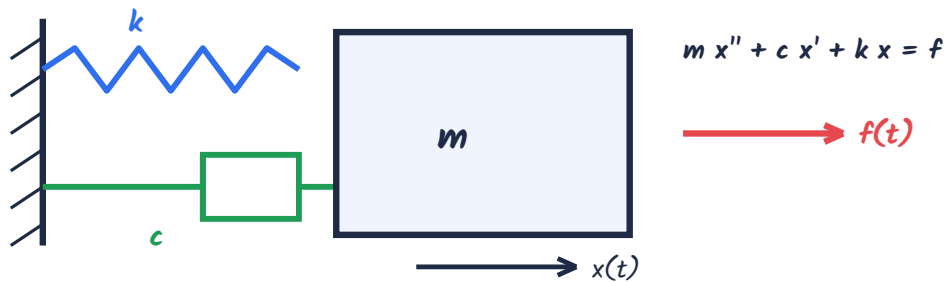
Sofia: 'Nobody pays for theory. They pay for the moment the test proves the FEM wrong BEFORE the prototype fails.'

(2.2) SDOF theory - the alphabet

$$m \ddot{x} + c \dot{x} + k x = f(t) \text{ (linear, time-invariant)}$$

Lumped mass m , linear spring $k x$, viscous damper $c \dot{x}$. [TRAP] All of Ch.2 assumes **LINEARITY**. Gaps, friction, loose joints break it - later chapters teach detecting that. For now: linear or nothing works.

my sketch: the SDOF machine



Free vibration trial $x = X e^{st}$ gives the **characteristic equation** and two roots = **poles**. Three damping cases exist; your career lives in **underdamped** ($z < 1$):

$$\text{poles: } s = -z \omega_n \pm j \omega_d, \omega_d = \omega_n \sqrt{1-z^2}$$

$$\omega_n = \sqrt{k/m}, z = c/cc, cc = 2 m \omega_n = 2 \sqrt{k m}$$

Pole anatomy - tattoo this:

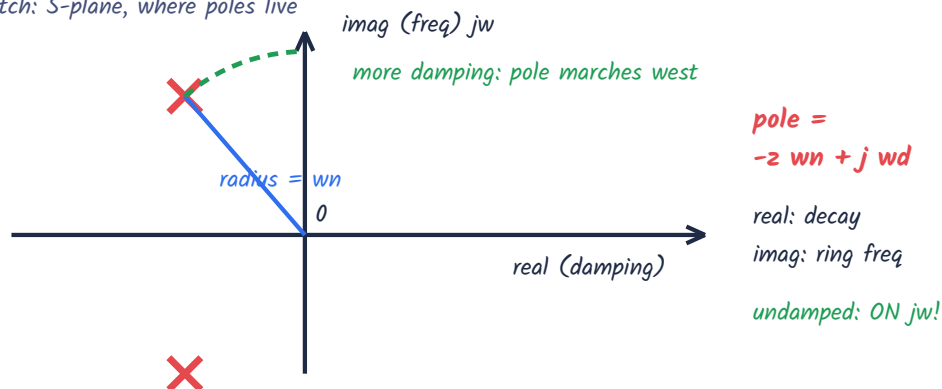
[KEY] Real part ($-z \omega_n$) = decay rate = damping. Imag part (ω_d) = ringing frequency. Poles come in **complex-conjugate pairs**. Light damping (under 10%): $\omega_d \sim \omega_n$. And ω_n itself does not care about damping!

EX1 - feel the numbers:

$m = 2 \text{ kg}, k = 8000 \text{ N/m}, c = 8$; $\omega_n = \sqrt{8000/2} = 63.2 \text{ rad/s} \rightarrow f_n \sim 10.1 \text{ Hz}$. $cc = 2 \cdot 2 \cdot 63.2 \sim 253$. $z = 8/253 \sim 0.032$ (3.2% - typical welded steel!). Poles: $-2 \pm j63.2$. Peak gain $Q \sim 1/(2z) \sim 16x$. THAT is why resonance breaks things.

(S-plane) Where poles live

my sketch: S-plane, where poles live



-> **Undamped:** poles ON the $j\omega$ axis (eternal sine - never real, shown for completeness).

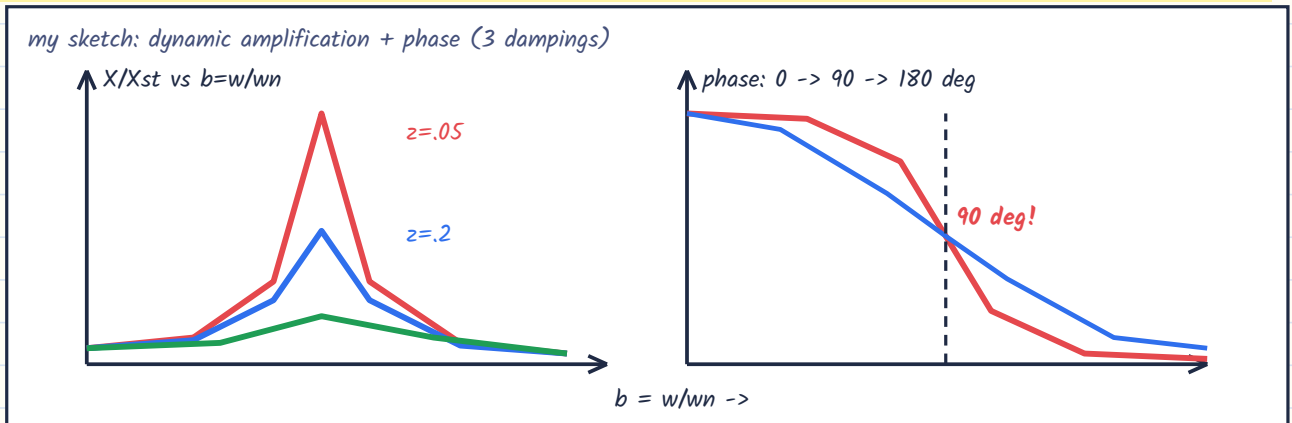
-> **More damping:** poles march **left** on a **circular arc**; ω_d shrinks. Critical: pair collides on the real axis.
Overdamped: splits into two real poles.

[KEY] Radius origin-to-pole = ω_n . Angle encodes z . **[LINK]** The S-plane is a MAP, poles are CITIES.

Damping pushes cities west; stiffness pushes them north. Your FRF is the view from the $j\omega$ highway.

(Forced) Amplification + the 90-degree signature

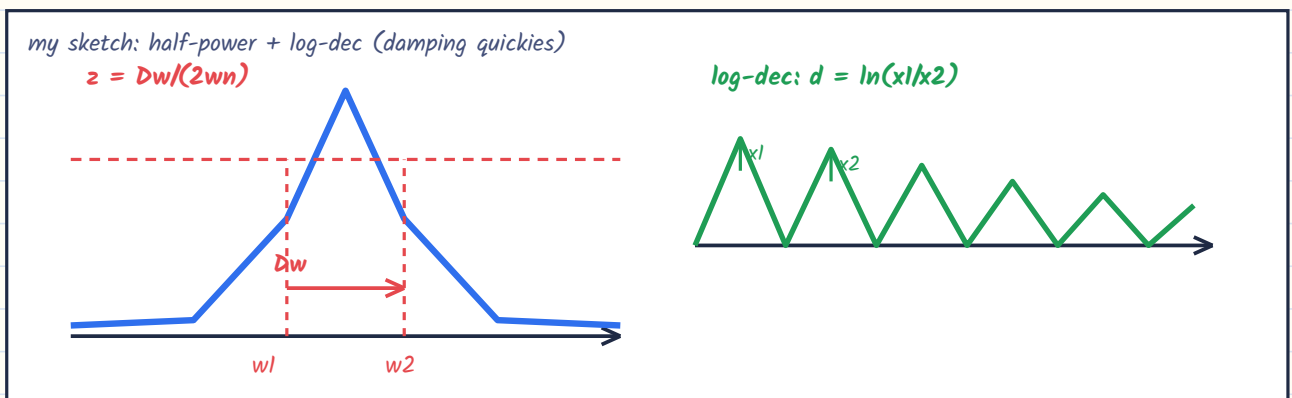
$$X/X_{static} = 1/\sqrt{(1-b^2)^2 + (2\zeta b)^2}, \quad b = \omega/\omega_n$$



-> Light damping: peak at $\omega_d \sim \omega_n$ with height $\sim 1/(2\zeta)$. Phase lag force->response: 0 deg -> **90 deg AT resonance** -> 180 deg.

[KEY] The **90-degree lag at resonance** is your most robust resonance detector - peaks can lie (noise, leakage), 90 deg rarely does.

(Damping) Half-power + log-dec: know, then outgrow



$$\text{HALF-POWER: } \zeta \sim (\omega_2 - \omega_1)/(2\omega_n) = D\omega/(2\omega_n)$$

$$\text{LOG-DEC: } d = \ln(x_1/x_2) \sim 2\pi\zeta$$

-> **Half-power (freq)**: peak/ $\sqrt{2}$ points ω_1, ω_2 . Wider peak = more damping. On IMAG part: half-height points; same points = REAL-part peaks; on Nyquist: **90 deg around the circle** from resonance.

-> **Log-dec (time)**: pluck it, watch ring-down over n cycles: $d = \ln(x_1/x_{n+1})/n$.

The book's honest verdict:

[TRAP] Historically important, exam-favorite - but real responses are almost never single-mode, so single-mode tricks mislead. **[LAB]** The professional answer: **modal parameter estimation**

(least-squares fit of poles + residues). Quickies for sanity checks and interviews; curvefitters for paychecks.

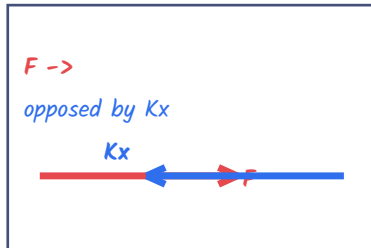
- Viktor

[TRY] EX2 - 10 seconds: peak 100 Hz, half-power at 99/101: $z \sim (101-99)/(2*100) = 1\%$. $Q \sim 50$. Lightly damped
- mind your leakage (Ch.1 flashback)!

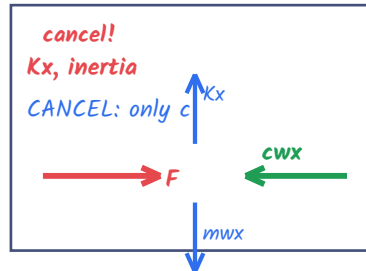
(Forces) Why resonance is a damping-only fight

my sketch: who balances F? (force vectors)

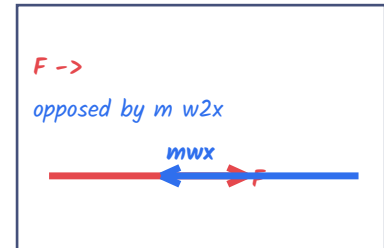
$w \ll w_n$: STATIC-like



$w = w_n$: DAMPING only!



$w \gg w_n$: MASS rules



Excitation	Balance	FRF region
$w \ll w_n$	$F \sim Kx$ (looks STATIC)	Stiffness-controlled
$w = w_n$	Kx and $m w^2 x$ CANCEL - only damping opposes F	Damping-controlled
$w \gg w_n$	$F \sim m w^2 x$ (inertia rules)	Mass-controlled

Deepest insight of 2.2:

[KEY] At resonance, *stiffness and inertia annihilate each other*; damping alone holds the bridge. That is why the peak measures damping and NOTHING else.

EX3 with numbers (EX1 at $w_n=63.2$, $X=1\text{mm}$): elastic $KX = 8\text{ N}$; inertia $m w^2 X = 8\text{ N}$ - equal, opposite, GONE. Damping $c w X \sim 0.5\text{ N}$ balances everything. 8 N vs 0.5 N : light damping = giant response.

(Laplace) One system, three costumes

- > **1. Polynomial:** $H(s) = 1/(m s^2 + c s + k)$ - straight from the equation.
- > **2. Pole-zero:** factored, roots exposed - poles = denominator roots.
- > **3. Partial fraction:** sum of residue/(s - pole) - screams 'ready for modal testing.'

[KEY] Same information, three outfits - NO new physics, just convenience. Inverse-Laplace the impulse: decaying exponentials $e^{(-z w_n t)} \sin(w_d t)$ - time-domain twin.

Residue = the number that carries the shape:

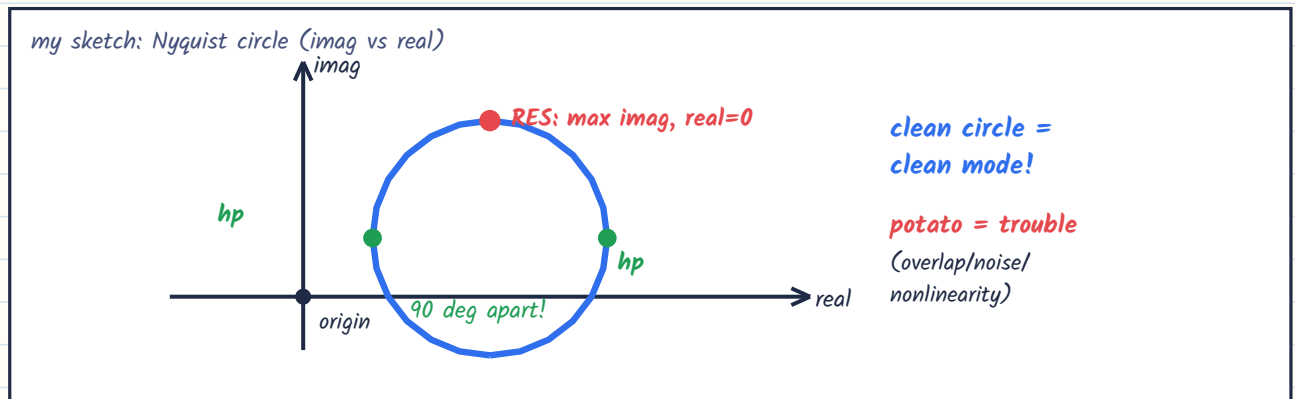
$H(s)$ explodes AT its poles (divide by zero) - the **residue theorem** extracts each pole's finite strength.

SDOF: a constant. MDOF: a MATRIX carrying every mode shape. **POLE + RESIDUE = complete DNA: rebuild the surface everywhere + the FRF at every f.**

(FRF) A slice of the surface at $s = jw$

Picture $H(s)$ as a tent over the S -plane with infinite spikes over each pole. The FRF is the *slice along the $j\omega$ axis* - walk it with a knife; the cut edge is your measurement. Near-axis poles (light damping) = tall sharp slices.

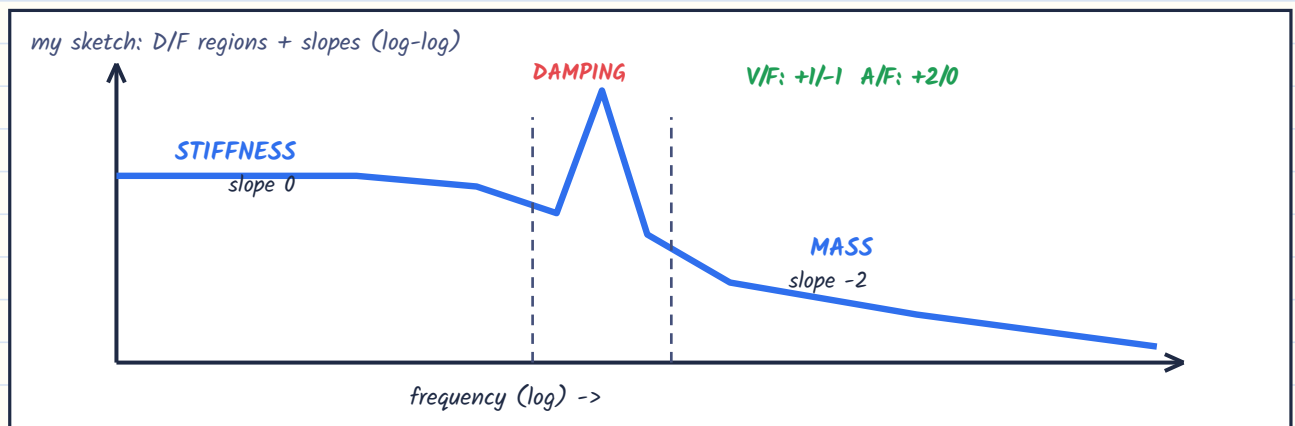
Portrait	Shows	Resonance reads as
Bode (mag + phase)	Peak + 90 deg, 180 total flip	Peak + 90 deg crossing
Co-quad (real + imag)	Real crosses ZERO, imag PEAKS	Imag peak, real = 0
Nyquist (imag vs real)	Near-CIRCLE	Opposite origin; hp pts 90 deg around



[LAB] Learn to love Nyquist. Clean circle = clean mode. A POTATO = overlapping modes, noise, or nonlinearity telling you to slow down. - Lena



(Forms) D/F, V/F, A/F: the slope detective kit



Name	Symbol	From D/F	Stiff slope	Mass slope
Compliance	D/F	-	0 (flat)	-2
Mobility	V/F	$\times j\omega$	+1	-1
Inertance	A/F	$\times (j\omega)^2 = -\omega^2$	+2	0 (flat)

[KEY] A/F = $-\omega^2$ (D/F): acceleration and displacement are **180 deg apart**, scaled by ω^2 . **[LAB] Detective trick:** unlabeled plot? Far-left/right slopes (0,-2)=D, (+1,-1)=V, (+2,0)=A. Catches instrument mix-ups in seconds.

[TRY] EX4 - name that plot: low end rises +2/decade, high end flat -> A/F acceleration - your everyday hammer + accelerometer FRF!

(2.3) MDOF theory - the main event

Coupled equations in matrix form (start: 2DOF):

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F\}$$

Mass-1's equation contains mass-2's motion and back - **coupled**. **[KEY]** Off-diagonals of **C** and **K** ARE the coupling. All matrices **square + symmetric** - the seed of reciprocity and of Ch.1's one-row/column rule. Linear + time-invariant, same honesty clause.

Eigensolution: frequencies + shapes fall out together:

$$([K] - \omega^2[M])\{u\} = 0 \rightarrow \det([K] - \omega^2[M]) = 0$$

- > **1. Determinant** = polynomial -> roots = eigenvalues (freqs squared). Small: Jacobi/Givens/Householder. Giant FEM: iterative, lowest modes only (subspace/Lanczos).
- > **2. Substitute** $\omega^2 = \omega^2$ back, solve (Crout/Cholesky/LDL) -> vector IS **mode shape 1**. Repeat per mode.
- > **3. Physics:** at each solution elastic = inertial forces: **dynamic equilibrium**, oscillating about **nodes**, equal +/- lobes. Mode 1 blue, mode 2 red - book colors kept!

[KEY] Frequency + vector = **EIGENPAIR**. Shapes are **linearly independent + orthogonal** wrt **M** and **K** - the miracle this chapter rides on.

EXS - 2DOF intuition:

Two floor masses, three springs. **Mode 1:** masses move TOGETHER (low f, middle spring lazy). **Mode 2:** masses move OPPOSITE (high f, middle spring works hard). Two independent dance moves; every real motion = a mix. That sentence scales to a million DOFs.

Orthogonality: the uncoupling keys:

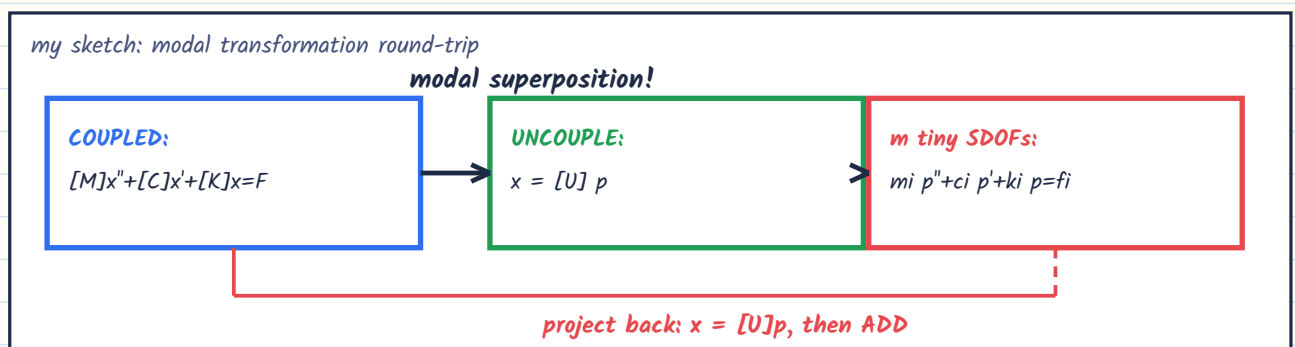
$$u_i(T)[M]u_j = 0, u_i(T)[K]u_j = 0 \quad (i \neq j)$$

$$u_i(T)[M]u_i = m_i \text{ (modal mass)}, u_i(T)[K]u_i = k_i$$

Stack shapes as columns -> **modal matrix [U]**. Change coordinates:

$$\{x\} = [U]\{p\} \rightarrow m_i \ddot{p} + c_i \dot{p} + k_i p = f_i$$

Modal force $f_i = u_i(T) \{F\}$: how hard **F** pushes mode **i**.



Why this is THE trick:

[KEY] Each mode becomes its own SDOF. Solve m tiny equations (often $m \ll n$: a handful of modes runs a million-DOF FEM!), project back with $[U]$, ADD. That round-trip = **modal superposition**, the most-used algorithm in structural dynamics.

Viktor: 'Test engineers walk it BACKWARDS: measure FRFs $\rightarrow$ extract poles/residues $\rightarrow$ animate shapes. Same bridge.'

Laplace for MDOF: global poles, shape-carrying residues:

$$[H(s)] = [B(s)]^{-1} = \text{adj}[B(s)] / \det[B(s)]$$

$\rightarrow$ System matrix $[B(s)]$ symmetric (M/C/K are) $\rightarrow$ **2n poles**, pairs per mode.

[KEY] Denominator $\rightarrow$ poles: **IDENTICAL** for every FRF (same roots wherever you measure) = **GLOBAL properties**. Numerator $\rightarrow$ residue matrix: symmetric, gives **reciprocity**, and at each pole vectors $\sim$ the **mode shapes** - EVERY row AND column valid!

[TRAP] Ch.1's node rule, birth certificate found: reference at a node zeroes that mode's residue across the whole row/column.

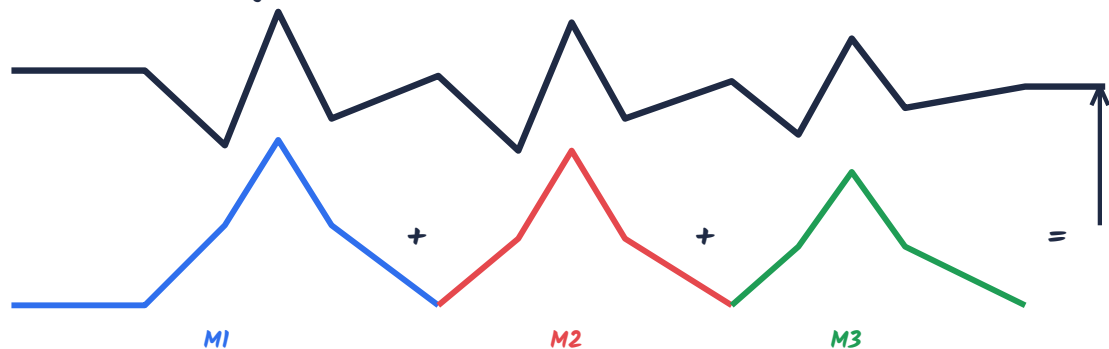
The FRF summation - two faces, one equation:

$$h_{ij}(\omega) = \sum_r A_{ijr} / (\omega^2 - \omega_r^2) \quad [+ \text{conj}] \quad (\text{residue face})$$

$$h_{ij}(\omega) = \sum_r u_{ir} u_{jr} / [m_r(\omega_r^2 - \omega^2)] \quad (\text{shape face})$$

Read the shape face like a story: each mode = its **SDOF amplifier** (pure 2.2!) $\times$ **uir** $\times$ **ujr** (output-shape $\times$ input-shape FILTERS). Weak shape at hammer OR sensor $\rightarrow$ mode hides in that FRF.

my sketch: FRF = sum of modal SDOF humps
SUMMED FRF (what you measure)



Sine at any f = SUM of modal responses: **below all modes** $\rightarrow$ mode 1 leads; **near mode 2** $\rightarrow$ mode 2 takes over; **between** $\rightarrow$ Ch.1's hybrid. Same moral, now with equations.

Cantilever, 3 points: the theory on one page:

FRF	Role	Watch for
h33 (drive, tip)	Sum of all modal oscillators; peak-valley alternation	Per-mode decomposition underneath
h32, h31 (cross)	Same poles, new residue weights	A mode shrinks as force nears its node
ROW (roving hammer)	Fixed accel reference	Ch.1 table: DERIVED now

FRF	Role	Watch for
COLUMN (fixed shaker)	Fixed force reference	Read mode 1 down column 3!

Imag-waterfall (tip ref) draws all three shapes in the air. **[KEY]** Any row/column works.

EX6 - estimate by hand:

Mode-1 imag peaks down column 3: [0.4, 1.1, 2.0] -> divide by tip: **[0.2, 0.55, 1.0]** - first-bending shape, root to tip. You just did modal parameter estimation with a pencil.

Time-frequency-modal: three doors, one room:



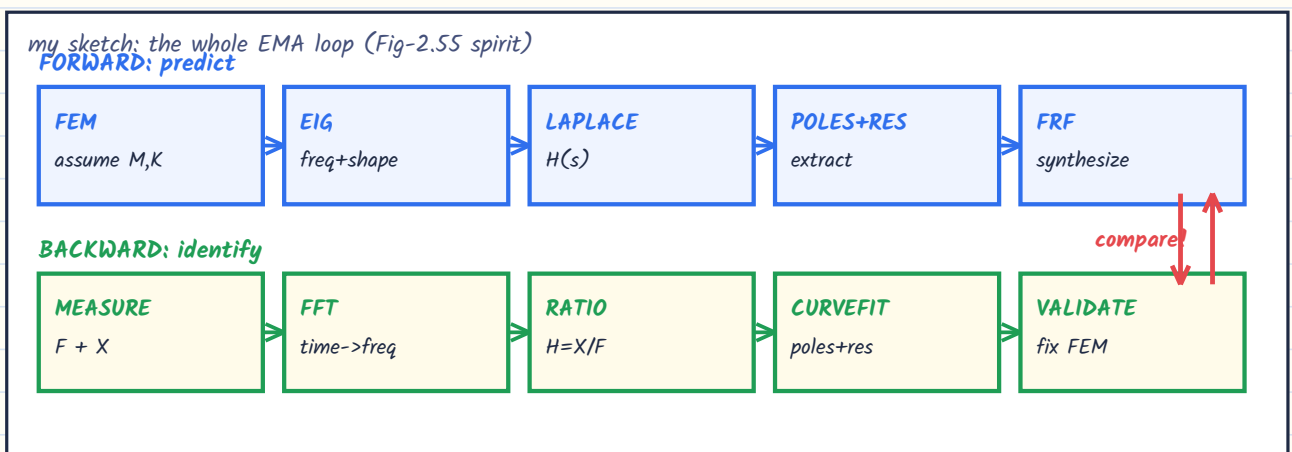
Time: tip = sum of decaying sines. **Freq:** FRF = sum of SDOF oscillators. **Modal:** decoupled SDOF engines + shapes. Convert freely - physics never changes, only your clothes do.

Turbine blade: the full computation (2.3.9):

Impulse near root (cantilever model), tip response wanted: eigensolve FEM -> modal $m/c/k$ + modal force per mode -> solve 3 trivial SDOF time responses -> **project back with shapes** -> add. Total tip trace with each mode's share beneath. **[KEY]** This is EVERY commercial solver's inner loop.

[TRY] EX7 - why truncation works: blade needs 0-50 Hz; modes at 8, 22, 47, 130, 260 Hz... keep 3 ($m=3 \ll n$); higher modes answer quasi-statically - hello, residuals (1.7 callback)!

(2.4 + Revision) The whole EMA loop + recap



-> **Forward (predict):** assume M, K -> eigensolve -> Laplace $H(s) = \text{adj}/\det$ -> poles + residues -> synthesize ANY FRF.

-> **Backward (identify):** measure $F+X$ -> FFT -> $H=X/F$ -> curvefit poles + residues -> validate FEM. SAME math objects, opposite direction!

[KEY] The gap = modeling error + measurement error. Your career = shrinking both. Next: **Ch.3** signals (sampling/leakage/windows), **Ch.4** excitation mastery, **Ch.5** the curvefitting zoo.

#	Big idea	Punchline
1	Pole anatomy	Real = damping, imag = freq; pairs
2	S-plane map	Damping pushes west on circle; $r = \omega_n$
3	Resonance signature	90 deg lag; K/x cancel; damping resists
4	Damping quickies	$z \sim D\omega/2\omega_n$; $d \sim 2\pi z$; pros fit
5	3 costumes	Polynomial / pole-zero / partial fractions
6	Residue	Pole strength; pole+residue = DNA
7	FRF = slice	$H(j\omega)$ cut along $j\omega$ axis
8	3 portraits	Bode / co-quad / Nyquist circle
9	Regions	Low stiff, reso damp, high mass
10	D/V/A slopes	(0,-2)/(+1,-1)/(+2,0): ID any plot
11	Eigensolution	det->freqs->shapes; equilib.+nodes
12	Uncouple	$x=[U]p$; m SDOFs replace n ($m \ll n$)
13	Global poles	Same denom; residues=shapes+reciprocity
14	FRF sum	$h_{ij} = \text{SUM shape-x-shape} \times \text{SDOF}$
15	Full loop	FEM->predict vs test->extract; same objects

$$\omega_n = \sqrt{k/m} \quad - \quad s = -z\omega_n \pm j\omega_d \quad - \quad z \sim D\omega/2\omega_n \quad - \quad ([K] - \omega^2[M])u = 0 \quad - \quad x = [U]p$$



(Self-test) 10 questions – no peeking!

- * Q1. Pole at $-3 \pm j40$. Give z , ω_n , ω_d . Light or heavy damping?
- * Q2. Why does the resonance peak measure damping and nothing else?
- * Q3. Half-power at 49.5/50.5 Hz, peak 50 Hz. Estimate z and Q .
- * Q4. Real part zero + imaginary peak at some frequency. What is it?
- * Q5. Nyquist looks like a potato, not a circle. List 3 suspects.
- * Q6. Unlabeled FRF: flat low end, -2 slope high end. Which form? Get V/F ?
- * Q7. Why are poles 'global' but residues 'local'?
- * Q8. Reference at a mode's node kills it – prove in one line from hij.
- * Q9. 200k-DOF FEM, response to 100 Hz, 12 modes below. How many equations? Rest?
- * Q10. FEM says 88 Hz, test says 82 Hz. One modeling + one test suspect?

Answers (really no peeking!):

- A1. $\omega_d=40$, $z \approx 3 \rightarrow \omega_n \approx 40.1$, $z \approx 7.5\%$ – moderate.
- A2. At ω_n , elastic/inertial forces cancel exactly; only $c \dot{x}$ opposes F .
- A3. $z \approx (50.5-49.5)/(2 \cdot 50) = 1\%$; $Q \approx 50$.
- A4. Resonance on a co-quad plot (clean, separated mode).
- A5. Overlapping modes, noise/leakage, nonlinearity (or calibration).
- A6. D/F compliance. Multiply by $j\omega$.
- A7. Poles = denominator roots, same all i,j ; residues = numerator, per pair.
- A8. $u_{jr} = 0$ at node $\rightarrow$ every $u_{ir} \times u_{jr} = 0$ for mode r . QED.
- A9. 12 modal SDOFs; higher modes quasi-static via residuals.
- A10. Modeling: wrong BCs / M-K layout. Test: mass loading, calibration, temperature.

Theory-to-lab checklist:

Before: predict band + mode count? reference off expected nodes? A/F or V/F – slope-check first plot?

During: 90 deg at every claimed resonance? Nyquist circles? drive-point alternation? coherence ~ 1 ?

After: poles stable across rows/columns (global!)? shapes orthogonal-ish? FEM updated + re-compared?

(Glossary) Ch.2 definitions

Pole - root of char. equation; $-z \omega_n \pm j \omega_d$; damping + freq.

Residue - pole's finite strength; MDOF: shape- i x shape- j .

S-plane - real-vs-imag pole map; $j\omega$ axis hosts FRFs.

Dynamic amplification - X/X_{static} vs b ; peak $\sim 1/(2\zeta) = Q$.

Half-power / log-dec - freq-/time-domain damping quickies.

Transfer function $H(s)$ - response/input in Laplace land; S-plane surface.

Partial fractions - $H(s)$ as SUM residue/(s -pole); test-ready form.

Bode / co-quad / Nyquist - mag-phase / re-im-vs-f / im-vs-re portraits.

Compliance/mobility/inertance - D/F, V/F, A/F (+ stiffness/impedance/mass inverses).

Eigenvalue/vector/pair - ω^2 / shape / the pair; $\det([K]-L[M])=0$.

Orthogonality - shapes decouple M and K; gives modal m, k.

Modal matrix $[U]$ - shapes as columns; $x=[U]p$ uncouples.

Modal force - $u(T)F$: how hard input pushes each mode.

Modal superposition - solve tiny modal SDOFs, project back, add.

System matrix $[B(s)]$ - s-domain coefficient matrix; symmetric.

Global property - same for all FRFs (poles, freqs, damping).

Truncation - keeping $m \ll n$ modes + residual compensation.

Council sign-off: Meera (theory) ok - Viktor (lab) ok - Lena (signals) ok - Arjun (pedagogy) ok - Sofia (field) ok.

Next stop: Ch.3 signals - sampling, leakage, windows!

